

Study Report: Python for Data Analysis - Data Analyst

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Family: Data Analyst
Level: Beginner
Chapters: 7
Source: Reference: Python for Data Analysis - Wes McKinney

Official sources and free libraries

- <https://wesmckinney.com/book/>

Summary

Study report for *Python for Data Analysis* by Wes McKinney (Beginner level) mapped to the Data Analyst role. pandas, NumPy, and data wrangling - the analyst's core toolkit.

Chapters

Track: Book-Intro

Chapter 1: About Python for Data Analysis [Beginner]

Chapter focus: About Python for Data Analysis (Level: Beginner)**

This study report summarizes *Python for Data Analysis* by Wes McKinney for the Data Analyst role. The resource is rated Beginner level. pandas, NumPy, and data wrangling - the analyst's core toolkit. Use this report alongside the official material to prepare for CodeQuest review.

Code Reference:

```
# Python for Data Analysis
# Author: Wes McKinney
# Role: Data Analyst
# Level: Beginner
```

What it shows: Metadata block records the source book and target role family.

Topic guide

What it actually is

A structured study report based on Python for Data Analysis.

When to use it

When learning Data Analyst skills at Beginner level.

Where to use it

Data Analyst training paths and certification prep.

Real use example

A learner reads this report before starting the full book or free online guide.

Track: Book-Context

Chapter 2: Why This Book Matters for Your Role [Beginner]

Chapter focus: Why This Book Matters for Your Role *(Level: Beginner)**

Data Analyst professionals use ideas from Python for Data Analysis to solve real workplace problems. pandas, NumPy, and data wrangling - the analyst's core toolkit. This chapter explains how the book fits into your learning path and what you should be able to do after studying it.

Code Reference:

Role: Data Analyst

Book focus: pandas, NumPy, and data wrangling - the analyst's core toolkit.

Recommended level: Beginner

What it shows: Connects the book topic to the job role outcomes.

Topic guide

What it actually is

Role-aligned learning connects theory to job tasks.

When to use it

During career planning and syllabus design.

Where to use it

Data Analyst bootcamps and CodeQuest teacher assignments.

Real use example

A teacher assigns this book report before a beginner skills checkpoint.

Track: Book-Topics

Chapter 3: Key Topics Covered [Beginner]

Chapter focus: Key Topics Covered *(Level: Beginner)**

The main topics in Python for Data Analysis include practical concepts described as: pandas, NumPy, and data wrangling - the analyst's core toolkit. Study each topic with hands-on practice. Take notes on definitions, workflows, and examples that match tools used in Data Analyst jobs today.

Code Reference:

- Topic 1: core concept
- Topic 2: core concept
- Topic 3: core concept
- Topic 4: core concept

What it shows: Topic list guides what to study chapter-by-chapter in the source.

Topic guide

What it actually is

Topic maps turn a book into actionable learning objectives.

When to use it

Before reading and while building chapter summaries.

Where to use it

Self-study, flipped classroom, and revision.

Real use example

A student maps each book chapter to a CodeQuest report section.

Track: Book-Applied

Chapter 4: Applied: Data Analyst Role in 2024-2026 [Beginner]

Chapter focus: Data Analyst Role in 2024-2026** *(Level: Beginner)

Data analysts turn raw business data into decisions. Modern teams expect SQL fluency, spreadsheet agility, Python/pandas for heavier work, and clear communication. You partner with product, marketing, and finance - not just build charts.

Code Reference:

```
SELECT COUNT(*) AS total_orders
FROM orders
WHERE order_date >= '2025-01-01';
```

What it shows: A simple count query answers 'how many orders this year?' - the bread-and-butter analyst task.

Topic guide

What it actually is

A data analyst queries, cleans, analyzes, and explains data to drive business action.

When to use it

When stakeholders need metrics, trends, or evidence for a decision.

Where to use it

SaaS dashboards, retail sales reviews, healthcare operations, and ed-tech reporting.

Real use example

A CodeQuest teacher asks 'which learning path has the highest completion rate?' - you answer with SQL and a one-slide chart.

Chapter 5: Applied: SQL SELECT, FILTER, and GROUP BY [Beginner]

Chapter focus: SQL SELECT, FILTER, and GROUP BY** *(Level: Beginner)

SQL is the analyst's primary language for warehouse data. SELECT picks columns; WHERE filters rows; GROUP BY aggregates by category. Always alias expressions and comment non-obvious business logic.

Code Reference:

```
SELECT region,
       COUNT(*) AS orders,
       SUM(amount) AS revenue
FROM sales
WHERE status = 'completed'
GROUP BY region
ORDER BY revenue DESC;
```

What it shows: GROUP BY region rolls up revenue per region; ORDER BY surfaces top performers first.

Topic guide

What it actually is

SQL is a declarative language for querying relational tables.

When to use it

Whenever data lives in PostgreSQL, BigQuery, Snowflake, or DuckDB.

Where to use it

Revenue reports, user counts, inventory snapshots, and audit extracts.

Real use example

Marketing wants top 5 regions by Q1 revenue - one GROUP BY query exports to Excel in seconds.

Chapter 6: Applied: Excel Tables, Pivot Tables, and XLOOKUP [Beginner]

Chapter focus: Excel Tables, Pivot Tables, and XLOOKUP** *(Level: Beginner)

Excel remains the universal analyst interface. Convert ranges to Tables for auto-expansion;

PivotTables summarize exports; XLOOKUP replaces brittle VLOOKUP. Use Power Query when files repeat weekly.

Code Reference:

```
=XLOOKUP(A2, Products[SKU], Products[Price], "Not found")
```

What it shows: XLOOKUP finds price by SKU with a clear not-found fallback.

Topic guide

What it actually is

Excel is a spreadsheet tool for exploration, ad-hoc analysis, and stakeholder handoffs.

When to use it

For quick what-if models, finance-friendly formats, and non-technical audiences.

Where to use it

Budget trackers, campaign ROI sheets, and executive one-pagers.

Real use example

Finance shares a CSV export - you build a pivot by department and month before the standup.

Track: Book-Plan

Chapter 7: Study Plan and Practice [Beginner]

Chapter focus: Study Plan and Practice *(Level: Beginner)**

Finish Python for Data Analysis with a weekly plan: read one section, write a summary, complete one exercise, and reflect on how it applies to Data Analyst. If a free edition exists, practice every example. Submit your notes and one mini-project demo for teacher review.

Code Reference:

```
Week 1: Read + notes  
Week 2: Exercises  
Week 3: Mini project  
Week 4: Review + quiz
```

What it shows: A 4-week plan turns reading into demonstrable skill.

Topic guide

What it actually is

Structured study plans improve retention and portfolio outcomes.

When to use it

After finishing the key topics chapters.

Where to use it

CodeQuest end-of-unit assessments.

Real use example

A learner completes the study plan and uploads a beginner project screenshot.